
PWDFT-DCU Manual v1.0

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Chapter 1: Introduction to PWDFT-DCU

PWDFT-DCU (Plane Wave Density Functional Theory) is a high-performance, parallel, and scalable software package for first-principles electronic structure calculations and *ab initio* molecular dynamics (AIMD) simulations. Based on the open-source PWDFT, it is specifically ported and optimized for the **Deep Computing Unit (DCU)** heterogeneous hardware on the Sugon advanced supercomputing platform.

By solving the Kohn-Sham equations, PWDFT-DCU enables the simulation of solid-state materials and molecular systems. It is comparable to software packages such as VASP and Quantum ESPRESSO. PWDFT is written in C/C++ and employs a CPU-MPI and DCU-HIP heterogeneous parallel framework to achieve extreme high-performance computing.

Key advantages of PWDFT-DCU include:

1. **Advanced First-Principles Development Platform:** PWDFT-DCU integrates state-of-the-art acceleration algorithms, including LOBPCG/PPCG eigensolvers, ACE, PCDIIS, and ISDF. These methods make large-scale hybrid density functional calculations up to **100 times faster** than existing algorithms.
2. **CPU-DCU Heterogeneous Acceleration:** Fully supports CPU-DCU heterogeneous acceleration. The DCU is a GPU-like graphic accelerator provided by Sugon. By leveraging DCU acceleration, PWDFT-DCU achieves significantly faster computational speeds.
3. **World-Leading Scalability:** Thanks to its exceptionally well-designed parallel architecture, PWDFT-DCU boasts top-tier scalability among plane-wave basis set software. The CPU version scales efficiently to **4,000 atoms using 8,000 CPU cores**, while the DCU version can seamlessly scale up to thousands of cards.

For more details on the methods and algorithms used in PWDFT-DCU, please refer to the following publications:

1. Wei Hu, Lin Lin, Amartya S Banerjee, Eugene Vecharynski, and Chao Yang. Adaptively compressed exchange operator for large-scale hybrid density functional calculations with applications to the adsorption of water on silicene. *J. Chem. Theory Comput.*, 13(3):1188–1198, 2017.

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2. Wei Hu, Lin Lin, and Chao Yang. Projected commutator diis method for accelerating hybrid functional electronic structure calculations. *J. Chem. Theory Comput.*, 13(11):5458–5467, 2017.
 3. Wei Hu, Lin Lin, and Chao Yang. Interpolative separable density fitting decomposition for accelerating hybrid density functional calculations with applications to defects in silicon. *J. Chem. Theory Comput.*, 13(11):5420–5431, 2017.
 4. Kun Dong, Wei Hu, and Lin Lin. Interpolative separable density fitting through centroidal voronoi tessellation with applications to hybrid functional electronic structure calculations. *J. Chem. Theory Comput.*, 14(3):1311–1320, 2018.
 5. Wei Hu, Jie Liu, Yingzhou Li, Zijing Ding, Chao Yang, and Jinlong Yang. Accelerating excitation energy computation in molecules and solids within linear-response time-dependent density functional theory via interpolative separable density fitting decomposition. *J. Chem. Theory Comput.*, **16**(2):964–973, 2020.
 6. Lingyun Wan, Xiaofeng Liu, Jie Liu, Xinming Qin, Wei Hu and Jinlong Yang. Hybrid MPI and OpenMP parallel implementation of large-scale linear-response time-dependent density functional theory with plane-wave basis set. *Electron. Struct.*, 3(2):024004, 2021.
 7. Junwei Feng, Lingyun Wan, Jielan Li, Shizhe Jiao, Xinhui Cui, Wei Hu, and Jinlong Yang. Massively parallel implementation of iterative eigensolvers in large-scale plane-wave density functional theory. *Comput. Phys. Commun.*, 299:109135, (2024).

Chapter 2: Compilation and Installation

2.1 Installation and Execution

PWDFT-DCU is primarily written in C/C++ and integrates MPI/OPENMP/HIP parallel tools. It relies on mathematical libraries such as FFTW, BLAS/LAPACK, ScaLAPACK, and LIBXC on the CPU side, as well as hipblas and hipfft on the DCU side.

2.1.1 Directory Structure

Download and extract the PWDFT-DCU package from GitHub:

```
git clone [https://github.com/HXTP/PWDFT-DCU]
tar -zxvf PWDFT-DCU.tar.gz
cd PWDFT-DCU
```

The main directory contains the following subdirectories: config, doc, examples, external, include, src, and utilities.

- config/: Example input files for compilation.
- src/: Core source program files.
- examples/: Source code for the main process and the generated executable.
- include/: Header files.
- external/: External libraries (lbfgs and qrqp).
- utilities/: MATLAB scripts for testing.

Key files include make.inc (the primary makefile), env.sh (compilation environment configuration), and compile.sh (one-click compilation script).

2.1.2 Compilation Method

(1) Prerequisites: PWDFT-DCU requires the installation of three main libraries: fftw-3.3.10, libxc-6.2.2, and yaml-0.8.0. Environments for intel, gcc, and dtk must also be properly set up. (DTK is the software stack for DCU, analogous to CUDA for GPUs).

Example compilation for FFTW (similar for LIBXC):

```
./configure --prefix=/public/home/hw45888792/yaoyf/soft/pwdft-lib/lib/fftw-  
gcc CC=gcc FC=gfortran MPICC=mpicc --enable-shared --enable-static --enable-mpi  
--enable-threads --enable-openmp  
make  
make install
```

Example compilation for YAML:

```
mkdir build  
cd build  
cmake -DCMAKE_INSTALL_PREFIX=/home/yaoyf/software/pwdft-  
lib/lib/yaml-cpp-0.8.0 ..  
make  
make install
```

(2) Configure Environment (env.sh): Modify env.sh to set paths for fftw, libxc, yaml, intel, and dtk.

Example:

```
DIST_DIR=$( cd -P -- "$(dirname -- "$0")" && pwd -P )  
export DGDFT_DIR=${DIST_DIR}  
export HIP_DIR=/public/home/sghpc_sdk/Linux_x86_64/25.6/dtk/dcc-2506  
export LIBXC_DIR=/public/home/acent8f17/code/libxc-6.2.2  
export FFTW_DIR=/public/home/acent8f17/code/fftw-3.3.10  
export  
MKL_ROOT=/public/home/sghpc_sdk/Linux_x86_64/25.6/compilers/intel/2021  
.3.0/mkl  
export YAML_DIR=/public/home/acent8f17/code/yaml-cpp-0.8.0
```

(3) Configure Build Settings (make.inc): Modify make.inc to verify all paths.

- Set USE_GPU = 1 for the **DCU version**.
- Set USE_GPU = 0 for the **CPU version**.

Example:

```
#!/usr/bin/bash
```

```

COMPILE_MODE = release
#COMPILE_MODE = debug
USE_COMPLEX = 0
USE_OPENMP = 0
USE_PROFILE = 0
USE_PEXSI = 0
USE_GPU = 1

ifeq (${COMPILE_MODE}, release)
    COMPILE_DEF = -DRELEASE
    # COMPILE_FLAG = -fast -w -D GPUDIRECT -no-ipo
    COMPILE_FLAG = -O3 -w
endif

ifeq (${COMPILE_MODE}, debug)
    COMPILE_DEF = -DDEBUG=1
    COMPILE_FLAG = -O2 -w
endif

ifeq (${USE_COMPLEX}, 1)
    SCALAR_DEF = -DCOMPLEXSCALAR
endif

HIP_DIR=/public/home/sghpc_sdk/Linux_x86_64/25.6/dtk/dcc-2506
LIBXC_DIR=/public/home/acents8f17/code/libxc-6.2.2
FFTW_DIR=/public/home/acents8f17/code/fftw-3.3.10
MKL_ROOT=/public/home/sghpc_sdk/Linux_x86_64/25.6/compilers/intel/2021.3.0/mkl
YAML_DIR=/public/home/acents8f17/code/yaml-cpp-0.8.0

ifeq (${USE_GPU}, 1)
    HIPCC = $(HIP_DIR)/hip/bin/hipcc
    HIP_LIB = -L$(HIP_DIR)/lib -lamdhip64 -lgalaxyhip -lhipblas -lrocfft -lhipfft -L$(HIP_DIR)/roclblas/lib -
lroclblas -L$(HIP_DIR)/rocsolver/lib -lrocsolver
    HIPCC_FLAG = -I$(HIP_DIR)/include -DGPU -I$(HIP_DIR)/hip/include -
I$(HIP_DIR)/hip/include/hip/hcc_detail/cuda -I ../include --gpu-max-threads-per-block=1024
    HIP_FLAG = -D GPU -I$(HIP_DIR)/include -I$(HIP_DIR)/hip/include -
I$(HIP_DIR)/hip/include/hip/hcc_detail/cuda -I ../include --gpu-max-threads-per-block=1024
endif

ifeq (${USE_OPENMP}, 1)
    OPENMP_DEF = -DOPENMP
    OPENMP_FLAG = -fopenmp
endif

ifeq (${USE_PROFILE}, 1)

```

```

    PROFILE_FLAG = -g -pg
endif

ifeq (${USE_PEXSI}, 1)
    PEXSI_DEF     = -DPEXSI

    PEXSI_DIR     =
    DSUPERLU_DIR  =
    METIS_DIR     =
    SCOTCH_DIR    =

    PEXSI_INCLUDE =
    PEXSI_SRC_LIB  =
    DSUPERLU_LIB  =
    SCOTCH_LIB    =
    METIS_LIB     =
    PEXSI_LIB     =
    PEXSI_LIB     =
endif

# Includes
DGDFT_INCLUDE = -I$(DGDFT_DIR)/include
LIBXC_INCLUDE = -I$(LIBXC_DIR)/include
FFTW_INCLUDE  = -I$(FFTW_DIR)/include
YAML_INCLUDE  = -I$(YAML_DIR)/include
#MFFT_INCLUDE = -I$(MFFT_DIR)/include
INCLUDES      = ${DGDFT_INCLUDE} ${PEXSI_INCLUDE} ${LIBXC_INCLUDE}
               ${BLOPEX_INCLUDE} ${FFTW_INCLUDE} ${MFFT_INCLUDE} ${YAML_INCLUDE}
               ${HIPCC_INC}

```

(4) **Compile:** Run the compilation script:

```
sh compile.sh
```

Upon successful compilation, the executable file `pwdft` will be generated in the `examples/` directory.

Chapter 3: Input and Output

3.1 Basic Input

The input format for PWDFT-DCU is similar to Quantum ESPRESSO. It requires two main types of input files: pseudopotentials and parameter inputs.

PWDFT-DCU supports both **HGH** and **ONCV** pseudopotentials (named by element + .upf for ASCII or .bin for binary). If the system contains multiple elements, a corresponding pseudopotential file must be provided for each. ONCV potentials can be downloaded from <http://www.mat-simresearch.com>.

The config.yaml file is the primary input file, defining critical parameters such as atomic types, structural information, and calculation methods. Below is an input example for Si64:

```
UPF_File:
- Si_ONCV_PBE-1.2.upf

Mixing_Variable: density
Mixing_Type: broyden
Mixing_MaxDim: 8
Mixing_StepLength: 0.6

Temperature: 0
SCF_Outer_MaxIter: 3
SCF_Outer_Tolerance: 1e-6
SCF_Phi_MaxIter: 3
SCF_Phi_Tolerance: 1e-6
SCF_Inner_MaxIter: 3
SCF_Inner_Tolerance: 1e-6
Eig_MaxIter: 5
Eig_Tolerance: 1e-12

Pseudo_Type: ONCV
Use_VLocal: 1
Use_Atom_Density: 1

XC_Type: XC_HYB_GGA_XC_HSE06
Hybrid_Mixing_Type: pcdiis
Hybrid_ACE: 1
```

PW_Solver: PPCG
ScaLAPACK_Block_Size: 32

Ecutoff_Wavefunction: 20
Density_Grid_Factor: 2.0

FFT_Number_Type: even

Output_Density: 0
Restart_Density: 0

POSCAR for PWDFT

Atom_Types_Num : 1
Atom_Type : [14]
Atom_Num : [64]
Super_Cell : [20.524998, 20.524998, 20.524998]
Atom_Red :
- [0.0000000000, 0.0000000000, 0.0000000000]
- [0.0000000000, 0.2500000000, 0.2500000000]
- [0.2500000000, 0.0000000000, 0.2500000000]
- [0.2500000000, 0.2500000000, 0.0000000000]
- [0.3750000000, 0.1250000000, 0.3750000000]
- [0.1250000000, 0.1250000000, 0.1250000000]
- [0.1250000000, 0.3750000000, 0.3750000000]
- [0.3750000000, 0.3750000000, 0.1250000000]
- [0.5000000000, 0.0000000000, 0.0000000000]
- [0.5000000000, 0.2500000000, 0.2500000000]
- [0.7500000000, 0.0000000000, 0.2500000000]
- [0.7500000000, 0.2500000000, 0.0000000000]
- [0.8750000000, 0.1250000000, 0.3750000000]
- [0.6250000000, 0.1250000000, 0.1250000000]
- [0.6250000000, 0.3750000000, 0.3750000000]
- [0.8750000000, 0.3750000000, 0.1250000000]
- [0.0000000000, 0.5000000000, 0.0000000000]
- [0.0000000000, 0.7500000000, 0.2500000000]
- [0.2500000000, 0.5000000000, 0.2500000000]
- [0.2500000000, 0.7500000000, 0.0000000000]
- [0.3750000000, 0.6250000000, 0.3750000000]
- [0.1250000000, 0.6250000000, 0.1250000000]
- [0.1250000000, 0.8750000000, 0.3750000000]
- [0.3750000000, 0.8750000000, 0.1250000000]

- [0.5000000000 , 0.5000000000 , 0.0000000000]
- [0.5000000000 , 0.7500000000 , 0.2500000000]
- [0.7500000000 , 0.5000000000 , 0.2500000000]
- [0.7500000000 , 0.7500000000 , 0.0000000000]
- [0.8750000000 , 0.6250000000 , 0.3750000000]
- [0.6250000000 , 0.6250000000 , 0.1250000000]
- [0.6250000000 , 0.8750000000 , 0.3750000000]
- [0.8750000000 , 0.8750000000 , 0.1250000000]
- [0.0000000000 , 0.0000000000 , 0.5000000000]
- [0.0000000000 , 0.2500000000 , 0.7500000000]
- [0.2500000000 , 0.0000000000 , 0.7500000000]
- [0.2500000000 , 0.2500000000 , 0.5000000000]
- [0.3750000000 , 0.1250000000 , 0.8750000000]
- [0.1250000000 , 0.1250000000 , 0.6250000000]
- [0.1250000000 , 0.3750000000 , 0.8750000000]
- [0.3750000000 , 0.3750000000 , 0.6250000000]
- [0.5000000000 , 0.0000000000 , 0.5000000000]
- [0.5000000000 , 0.2500000000 , 0.7500000000]
- [0.7500000000 , 0.0000000000 , 0.7500000000]
- [0.7500000000 , 0.2500000000 , 0.5000000000]
- [0.8750000000 , 0.1250000000 , 0.8750000000]
- [0.6250000000 , 0.1250000000 , 0.6250000000]
- [0.6250000000 , 0.3750000000 , 0.8750000000]
- [0.8750000000 , 0.3750000000 , 0.6250000000]
- [0.0000000000 , 0.5000000000 , 0.5000000000]
- [0.0000000000 , 0.7500000000 , 0.7500000000]
- [0.2500000000 , 0.5000000000 , 0.7500000000]
- [0.2500000000 , 0.7500000000 , 0.5000000000]
- [0.3750000000 , 0.6250000000 , 0.8750000000]
- [0.1250000000 , 0.6250000000 , 0.6250000000]
- [0.1250000000 , 0.8750000000 , 0.8750000000]
- [0.3750000000 , 0.8750000000 , 0.6250000000]
- [0.5000000000 , 0.5000000000 , 0.5000000000]
- [0.5000000000 , 0.7500000000 , 0.7500000000]
- [0.7500000000 , 0.5000000000 , 0.7500000000]
- [0.7500000000 , 0.7500000000 , 0.5000000000]
- [0.8750000000 , 0.6250000000 , 0.8750000000]
- [0.6250000000 , 0.6250000000 , 0.6250000000]
- [0.6250000000 , 0.8750000000 , 0.8750000000]
- [0.8750000000 , 0.8750000000 , 0.6250000000]

3.2 Basic Output

The primary output file is statfile.0. If the calculation terminates normally, a binary file named STRUCTURE will also be generated.

statfile.0 contains a detailed log of the entire calculation process, including:

- Default and user-defined parameters (Common info & PWDFT info).
- Time statistics and energies for each SCF iteration step (Outer SCF inter #).
- Converged final energy (Energy section).
- Atomic forces (Atomic Force section).

Below is an output example for Si64:

```
*****
Common information
*****

Super cell          = +2.052499800000e+01 +2.052499800000e+01 +2.052499800000e+01
Grid Wavefunction   = 42          42          42
Grid Density        = 84          84          84
Mixing dimension     = 8

*****
PWDFT information
*****

SCF Phi MaxIter      = 3
SCF Phi Tol          = +1.000000000000e-06
Hybrid ACE           = true
Hybrid DF            = false
Hybrid Active Init    = false
Hybrid Mixing Type    = pcdiis
Hybrid DF Num Mu      = +6.000000000000e+00
Hybrid DF Num GaussianRandom = +2.000000000000e+00
Hybrid DF Tolerance   = +1.000000000000e-20
EXX div type         = 1
Subblock size (sbSize) in PPCG = 1

...

*****
SCF iteration # 1
*****
```

```

Time for iterSpinor   = 11  timeSpinor   = 0.504003051261
Time for iterGemmT    = 6933 timeGemmT    = 0.144381820501
Time for iterGemmN    = 3609 timeGemmN    = 0.0300053026876
Time for iterTrsm     = 11  timeTrsm     = 0.267780302995
Time for iterPotrf    = 6   timePotrf    = 0.00127897462517
Time for iterSyevd    = 1   timeSyevd    = 0.00347632415651
Time for iterSygvd    = 640 timeSygvd    = 0.00849758015829
Time for iterCPU2DCUCopy = 1823 timeCPU2DCUCopy = 1.48847118633
Time for iterDCU2DCUCopy = 3591 timeDCU2DCUCopy = 0.00776964092802
Time for iterOther    = 3119 timeOther    = 0.0296624231123
Time for PPCG in PWDFT is 2.66313773295

...

*****

Phi iteration # 1

*****

...

Fock energy   = -1.342311179049e+01[au]
Etot(with fock) = -2.519516085680e+02[au]
Efree(with fock) = -2.519516085680e+02[au]
dExx         = +3.513488274724e-03[au]

...

*****

Energy

*****

...

! EfreeHarris   = -2.656034364736e+02[au]
! Etot          = -2.522201839460e+02[au]
! Exc           = -6.399161877077e+01[au]
! Exx           = -1.345643581859e+01[au]
! Efree         = -2.522201839460e+02[au]
! Evdw          = +0.000000000000e+00[au]
! Eext          = +0.000000000000e+00[au]
! Fermi         = +1.851082068245e-01[au]
! HOMO          = +5.037050913203e+00[eV]

*****

Atomic Force

*****

atom 0  force +3.529712801794e-03 +3.392529926468e-03 -2.501753616226e-03
atom 1  force +3.359913344355e-03 -2.525220230981e-03 +3.539289634218e-03
atom 2  force +3.156341985853e-03 +2.095980554452e-03 -6.887703426064e-03
atom 3  force +7.023139827346e-03 -1.999685293601e-03 -3.196588376870e-03

```

...

atom 61 force -4.294879490025e-03 -2.209510741299e-04 -1.881297909926e-03

atom 62 force -1.048156718398e-02 -2.468083479364e-03 -3.015608991940e-03

atom 63 force -1.548901306710e-03 +4.257223915775e-03 -7.494820743794e-03

force for centroid : +4.270745637889e-03 -1.314826052659e-04 -5.702379961978e-03

Max force magnitude : +1.118251243172e-02

Output the structure information

! Total time for the SCF iteration = 75.2125261536 [s]

Chapter 4: Input Parameters

4.1 Basic Input Parameters

1. **Mixing_Variable**(String): The mixing variable for SCF calculations. Options: potential or density
Defalut: potential
2. **Mixing_Type**(String): The mixing scheme. Options: anderson or kerker+anderson.
Defalut: anderson
3. **Mixing_StepLength**(Float): The step length for mixing.
Defalut: 0.8
4. **Mixing_MaxDim**(Integer): The subspace dimension for mixing.
Defalut: 9
5. **Output_Density**(Integer): Whether to output the electron density.
Defalut: 0
6. **Restart_Density**(Integer): Whether to read an existing density.
Defalut: 0
7. **Temperature**(Float): Electronic temperature in Kelvin.
Defalut: 300.0
8. **Ecut_Wavefunction**(Float): Wavefunction cutoff energy in Hartree.
Defalut: 40.0

9. **Density_Grid_Factor**(Float): The grid expansion factor for electron density.

Defalut: 2.0

10. **PeriodTable**(String): Pseudopotential filename when using HGH type. (Not required for ONCV, where paths are defined within begin UPF_File ... end UPF_File).

11. **Pseudo_Type**(String): Pseudopotential type. Options: HGH and ONCV.

Defalut: HGH

12. **XC_Type**(String): Exchange-correlation functional. Options include XC_LDA_XC_TETER93, XC_GGA_XC_PBE, XC_HYB_GGA_XC_PBEH, XC_HYB_GGA_XC_HSE06.

Defalut: XC_LDA_XC_TETER93

13. **VDW_Type**(String): van der Waals correction type. Currently supports DFT-D2.

Defalut: None

14. **Super_Cell**(Float): Defines lattice parameters. PWDFT-DCU currently only supports orthogonal lattices.

15. **Atom_Types_Num**(Integer): Number of atomic species.

Defalut: 0

16. **Atom_Type**(Integer): Atomic numbers representing the species.

Defalut: 0

17. **Atom_Red**(Float): Defines fractional atomic coordinates.

4.2 Iteration Parameters

1. **SCF_Outer_MinIter**(Integer): Minimum outer SCF iterations.
Defalut: 3
2. **SCF_Outer_MaxIter**(Integer): Maximum outer SCF iterations.
Defalut: 30
3. **SCF_Outer_Tolerance**(Float): Convergence tolerance for outer SCF.
Defalut: 1e-6
4. **SCF_Inner_MinIter**(Integer): Minimum inner SCF iterations.
Defalut: 1
5. **SCF_Inner_MaxIter**(Integer): Maximum inner SCF iterations.
Defalut: 1
6. **Calculate_Force_Each_SCF**(Integer): Whether to compute forces at each SCF step.
Defalut: 0
7. **Eig_MaxIter**(Integer): Maximum eigensolver iterations within SCF.
Defalut: 3
8. **Eig_Tolerance**(Float): Convergence tolerance for the eigensolver.
Defalut: 1e-20
9. **PW_Solver**(String): Eigensolver algorithm. Options: PPCG and PPCGScaLAPACK.
10. **ScaLAPACK_Block_Size** (Integer): Block size for ScaLAPACK.

Defalut: 32

4.3 MD Input Parameters

1. **Ion_Move(String)**: Defines ion dynamics. Structural optimization options: bb, pgbb, nlcg, lbfgs, fire. MD options: verlet, nosehoover1, langevin.
Defalut: 40
2. **MD_Time_Step(Integer)**: Molecular dynamics time step.
Defalut: 0
3. **Ion_Max_Iter(Integer)**: Maximum number of ionic steps.
Defalut: 300.0
4. **Ion_Temperature(Float)**: Ionic temperature.
Defalut: linear
5. **MD_Extrapolation_Type (String)**: MD extrapolation type.
Defalut: density
6. **MD_Extrapolation_Variable (String)**: Variable for MD extrapolation. Options: density and wavefun.
Defalut: density

4.4 Hybrid Functional Parameters

1. **Hybrid_ACE(Integer)**: Whether to use the Adaptively Compressed Exchange (ACE) operator. Highly recommended for HSE acceleration.
Defalut: 1
2. **Hybrid_DF(Integer)**: Whether to use Density Fitting (ISDF algorithm).
Defalut: 0

-
3. **Hybrid_DF_Type(String)**: Density fitting method. Options: QRCP, Kmeans, Kmeans+QRCP.
Defalut: QRCP
 4. **Hybrid_DF_Num_Mu(Float)**: The Mu value used in density fitting.
Defalut: 6.0
 5. **Hybrid_DF_Num_GaussianRandom(Float)**: The size of the random Gaussian matrix for density fitting.
Defalut: 2.0

4.5 LR-TDDFT Input Parameters

1. **LRTDDFT(Integer)**: Whether to perform LR-TDDFT calculations.
Defalut: 0
2. **Extra_States(Integer)**: Number of empty states computed during SCF.
Defalut: 0
3. **Nvband(Integer)**: Number of occupied bands required for LR-TDDFT.
Defalut: 1
4. **Ncband(Integer)**: Number of unoccupied bands required for LR-TDDFT.
Defalut: 1
5. **Output_Excitation_Energy(Integer)**: Whether to output excitation energies.
Defalut: 0

Chapter 5: PWDFT-DCU Benchmark

Examples

A benchmark case for **Si64** (both CPU and DCU versions) is provided in the test/ directory.

- config.yaml: The input parameter file.
- Si_ONCV_PBE-1.2.upf: The pseudopotential file.
- statfile.0: The reference output file.
- srun.sh & intelmpi_bind.sh: Slurm job submission scripts for the Sugon supercomputing platform.

PWDFT-DCU is designed primarily to run on Sugon's advanced supercomputing platforms. The hardware environment typically consists of **Sugon C86 7185 32-core CPUs** and **8 Sugon DCU cards per node**. The software environment utilizes compiler/devtoolset/7.3.1, compiler/dtk/25.6, and compiler/intel/2021.3.0.

```
#!/bin/bash
#SBATCH -J c-p-a
#SBATCH -p newlarge
#SBATCH -N 1
#SBATCH --ntasks-per-node=8
#SBATCH --cpus-per-task=1
#SBATCH --gres=dcu:8
#SBATCH -o aimd-%j.log
#SBATCH -e aimd-%j.err
#SBATCH --exclusive

module purge
#module load sghpc-mpi-clang-mlnx/25.6
#module add sghpc-mpi-intel/25.6 mpi/intelmpi/2021.14.0
module use /public/software/modules/
module load compiler/intel/2021.3.0
#module load mpi/intelmpi/2021.14.0
module load mpi/intel/2021.3.0
module load sghpc-mathlib/25.6-intel
source /public/home/sghpc_sdk/Linux_x86_64/25.6/dtk/dcc-2506/env.sh
which hipcc
which mpirun
```

Multi-node execution

```
#mpirun -np 8 -genv UCX_TLS=rc,sm -genv FI_MPI_FABRICS ofi -genv FI_PROVIDER  
ucx ./intelpi_bind.sh /public/home/acent8f17/soft/dghf-DCU-0901/examples/pwdft
```

Single-node execution

```
mpirun -np 1 -genv UCX_TLS=rc,sm -genv FI_MPI_FABRICS ofi -genv FI_PROVIDER ucx -genv  
UCX_TLS=sm ./intelpi_bind.sh /public/home/acent8f17/yaoyf/software/dghf-CPU-2026.1.10/examples/pwdft
```

Script Parameter Explanations:

- -N: Specifies the number of nodes.
- -n: Specifies the total number of processes.
- --ntasks-per-node: Specifies the number of processes per node.
- -p: Specifies the partition/queue name for submission.
- --gres=dcu:8: Specifies the number of DCU cards utilized per node (8 in this example).
- Make sure to specify the absolute compiled path for pwdft when running the mpirun command.